

. External Libraries & CDN Dependencies

#	Library	Version	CDN URL	Purpose
1	Tailwind CSS	Latest (CDN build)	https://cdn.tailwindcss.com?plugins=forms,container-queries	Utility-first CSS framework with forms & container-queries plugins. Custom dark theme via tailwind.config
2	Three.js	r128	https://cdn.jsdelivr.net/npm/three.js/r128/three.min.js	3D WebGL rendering engine for the voice-reactive glass orb sphere
3	jsPDF	2.5.1	https://cdn.jsdelivr.net/npm/jspdf/2.5.1/jspdf.umd.min.js	Client-side PDF generation for analysis reports
4	jsPDF-AutoTable	3.8.2	https://cdn.jsdelivr.net/npm/jspdf-autotable/3.8.2/jspdf.plugin.autotable.min.js	Structured table rendering within jsPDF documents

5	ONNX Runtime Web	1.29.0	https://cdn.jsdelivr.net/npm/onnxruntime-web@1.29.0/dist/ort.min.js	WebAssembly-based ML inference engine for running Silero VAD ONNX model in-browser
6	Google Fonts — Inter	Variable (100–900)	https://fonts.googleapis.com/css2?family=Inter:wght@100..900	Primary typography (body, headings, labels, monospace UI)
7	Material Symbols Outlined	Variable weight/fill	https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined	Icon system for all UI buttons and status indicators

2. AI / ML Models

#	Model	Format	Source URL	Purpose
1	Silero VAD v5	ONNX	https://huggingface.co/runanywhere/silero-vad-v5/resolve/main/silero_vad.onnx	Voice Activity Detection — classifies audio frames as speech or silence

Silero VAD v5 Tensor Specification:

Direction	Tensor Name	Data Type	Shape	Notes
Input	input	Float32	[1, 576]	512 new samples + 64 rolling context
Input	state	Float32	[2, 1, 128]	LSTM hidden + cell state

Input	sr	Int64	[1]	Sample rate (16000n)
Output	output	Float32	[1, 1]	Speech probability [0..1]
Output	stateN	Float32	[2, 1, 128]	Updated LSTM state

3. Web Platform APIs Used

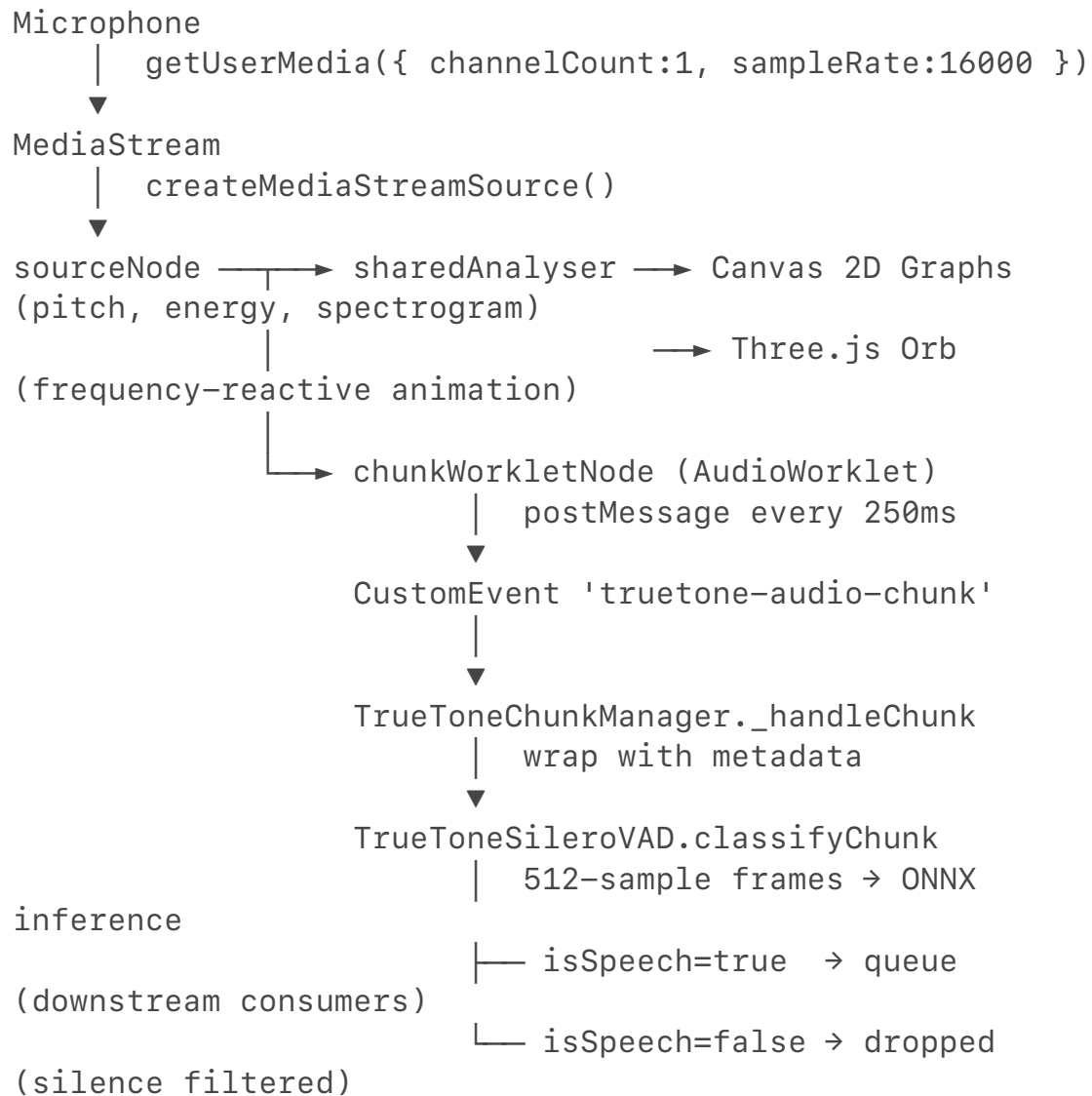
#	API	Usage	Location
1	Web Audio API — AudioContext	Audio graph management, sample rate control (16kHz target)	TrueToneAudio.startAnalysis()
2	Web Audio API — AudioWorklet	Real-time PCM chunk capture on audio rendering thread	audio-chunk-processor.js + _initChunkWorklet()
3	Web Audio API — AnalyserNode	FFT frequency data for pitch, energy, and spectrogram visualizations	sharedAnalyser (fftSize=2048)
4	Web Audio API — MediaStreamSource	Bridge from microphone MediaStream into Web Audio graph	createMediaStreamSource()
5	MediaDevices.getUserMedia()	Microphone capture (mono 16kHz preferred, stereo fallback)	navigator.mediaDevices.getUserMedia()
6	Permissions API	Pre-check microphone permission status without prompting	navigator.permissions.query({ name: 'microphone' })

7	Canvas 2D API	Pitch contour, energy waveform, and spectrogram graph rendering	graph-pitch, graph-energy, graph-spectrogram canvases
8	WebGL API (via Three.js)	3D voice-reactive glass orb with custom GLSL shaders	THREE.WebGLRenderer, THREE.ShaderMaterial
9	ResizeObserver	Responsive canvas/WebGL resizing when container dimensions change	Orb container + graph canvases
10	requestAnimationFrame	Animation loop for both 3D orb and 2D graph rendering	Orb animate() + graph renderLoop()
11	Performance API	High-resolution timestamps for audio chunk receipt time	performance.now() in chunk metadata
12	CustomEvent API	truetone-audio-chunk event bus between worklet and chunk manager	window.dispatchEvent(new CustomEvent(...))
13	LocalStorage	Persistent session history storage	Key: truetone_analysis_history_v2
14	Visibility API	Pause/resume audio context when tab is hidden	document.addEventListener('visibilitychange', ...)
15	Secure Context check	Guard AudioWorklet + model loading (HTTPS/localhost required)	window.isSecureContext

4. Custom JavaScript Modules (Global Objects)

#	Module	Global	Purpose
1	TrueToneAudio	window.TrueToneAudio	Centralized audio pipeline — mic capture, AudioContext, sourceNode, sharedAnalyser, AudioWorklet node
2	TrueToneChunkManager	window.TrueToneChunkManager	Receives raw PCM chunks, wraps with metadata (ID, timestamps, sample info), maintains in-memory queue
3	TrueToneSileroVAD	window.TrueToneSileroVAD	ONNX-based voice activity detection — classifies chunks, filters silence
4	TrueToneSessionClock	window.TrueToneSessionClock	Dynamic session timer displayed on the dashboard
5	TrueToneState	window.TrueToneState	Lightweight global state flags (analysisActive, etc.)
6	HistoryStore	(module-scoped)	LocalStorage adapter for session history CRUD
7	VoxGuard Orb	(IIFE-scoped)	Three.js 3D voice-reactive glass sphere with custom GLSL shaders

5. Audio Pipeline Architecture



6. UI Buttons — Complete Inventory

6.1 Login Screen

#	Button ID	Label	Action	API Endpoint
1	login-submit-btn	SIGN IN	Validates email + password fields, calls <code>authenticate User()</code> , transitions to start screen on success	Stub: POST / api/auth/ login (currently simulated with <code>setTimeout</code> — see TODO at line 3856)

2	login-toggle-pw	 (visibility icon)	Toggles password field between type="password" and type="text"	<i>Client-side only — no API</i>
3	login-forgot-link	Forgot Password?	Anchor link () — currently non-functional placeholder	Planned: POST /api/auth/forgot-password

6.2 Start / Landing Screen

#	Button ID	Label	Action	API Endpoint
4	start-analysis-btn	START ANALYSIS	Requests mic permission via getUserMedia(), creates AudioContext at 16kHz, loads AudioWorklet, loads Silero VAD ONNX model, navigates to analysis dashboard	<i>Client-side Web APIs only</i> (getUserMedia, AudioWorklet, ONNX Runtime)
5	sidebar-new-analysis-btn	New Analysis	Returns to main landing screen, calls returnToMainScreen()	<i>Client-side navigation only — no API</i>

6.3 Analysis Dashboard

#	Button ID	Label	Action	API Endpoint
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6	end-analysis-btn	Complete Analysis	Ends live session: stops mic tracks, closes AudioContext, generates session report, saves to localStorage, navigates to report view	<i>Client-side only — session data persisted to localStorage key truetone_analysis_history_v2</i>
7	audio-status-badge	Waiting for audio input	Click-to-connect mic badge: triggers TrueToneAudio.startAnalysis() if not already active, unlocks AudioContext on user gesture	<i>Client-side — triggers getUserMedia flow</i>

6.4 Report / History View

#	Button ID	Label	Action	API Endpoint
8	back-to-main-btn	← Back to Main Screen	Navigates back to landing screen via returnToMainScreen()	<i>Client-side navigation only — no API</i>

9	download-report-btn	DOWNLOAD REPORT	Generates and downloads PDF report for the selected session using jsPDF + jsPDF-AutoTable	<i>Client-side PDF generation — no API</i>
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6.5 History Sidebar (Dynamic Items)

#	Element	Label	Action	API Endpoint
10	Session list items	<i>(dynamic per session)</i>	Click loads the selected session's report into the report view panel	<i>Client-side — reads from localStorage</i>

7. API Endpoints Summary

IMPORTANT

The application currently runs **entirely client-side** with no backend API calls. All "API endpoints" below are either **stubbed** (with TODO comments in code) or **planned** for future implementation.

7.1 Authentication (Stubbed)

Method	Endpoint	Status	Notes
POST	/api/auth/login	 Stubbed	Currently simulated with 1.5s delay returning { success: true }. See code.html:3855-3871
POST	/api/auth/forgot-password	 Not Implemented	"Forgot Password?" link exists but is non-functional

7.2 External Model Endpoints (Active)

Method	URL	Status	Notes
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GET	https://huggingface.co/runanywhere/silero-vad-v5/resolve/main/silero_vad.onnx	 Active	Silero VAD model download (loaded by ONNX Runtime Web at mic activation)
GET	https://cdn.jsdelivr.net/npm/onnxruntime-web@1.29.0/dist/	 Active	WASM backend files for ONNX Runtime (auto-resolved via ort.env.wasm.wasmPaths)

7.3 Planned / Future API Points

Method	Endpoint	Purpose
POST	/api/analysis/upload	Upload speech chunks for server-side deepfake detection (not yet implemented)
GET	/api/sessions	Retrieve analysis session history from server (currently in localStorage)
POST	/api/sessions	Save analysis session to server backend
GET	/api/sessions/:id/report	Retrieve PDF report for a specific session

8. Data Storage

Storage	Key / Location	Data	Format
localStorage	truetone_analysis_history_v2	Array of completed analysis session objects	JSON string
In-Memory	TrueToneChunkManager._queue	Live audio chunk metadata queue (speech-only after VAD)	Array of JS objects
In-Memory	TrueToneSileroVAD._lstmState	Persistent LSTM state between VAD frames	Float32Array(256)

In-Memory	TrueToneSileroVA D._contextTail	64-sample rolling context for Silero's 576- sample input	Float32Array(64)
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9. Rendering & Visualization Stack

Component	Technology	Canvas ID / Container	Data Source
Voice-Reactive 3D Orb	Three.js r128 + custom GLSL vertex/fragment shaders	#voice-sphere- slot	sharedAnalyser FFT frequency data
Pitch / FO Contour	Canvas 2D API	#graph-pitch	sharedAnalyser frequency bin analysis
Energy Waveform	Canvas 2D API	#graph-energy	sharedAnalyser time-domain data
Spectrogram	Canvas 2D API + off-screen canvas scroll	#graph- spectrogram	sharedAnalyser frequency data (color-mapped)
PDF Report	jsPDF + jsPDF- AutoTable	(<i>download trigger</i>)	Session data from HistoryStore

10. Design System

Token	Value
Font Family	Inter (100–900 weight)
Icon Set	Material Symbols Outlined (variable weight/fill)
Color Mode	Dark mode only (html.dark)
Background	#121317 (surface), #08090C (report view)
Accent Primary	Amber-500 (#F59E0B) — gradients, CTAs, highlights
Accent Success	Emerald-400 — mic connected, session active
Accent Danger	Rose-500 — errors, end call, mic denied
Accent Neutral	#8E8E93 — secondary text, labels
Glass Effect	backdrop-blur-md + white/[0.04] bg overlays

Border Radius	rounded-xl (12px) for buttons, rounded-2xl (16px) for cards
Reduced Motion	prefers-reduced-motion media query respected globally

11. Security Requirements

Requirement	Implementation
Secure Context	AudioWorklet and ONNX model loading require HTTPS or localhost. Explicit window.isSecureContext guard with error logging.
Microphone Permission	Graceful error handling for NotAllowedError, NotFoundError, NotReadableError with user-friendly messages
No Hardcoded Credentials	Login form uses client-side validation; authenticateUser() is a stub awaiting backend API
Audio Privacy	Microphone tracks are explicitly stopped on session end / page unload via beforeunload handler